Intervals and absolute value equations

Two-vertex certification of nonsingular sign regularity

Difficulty: challenging **Importance:** interesting to specialist

Status: Lean verified

Area: structured matrices and interval linear algebra

Rating rationale: These ratings are historical and refer to the original question: extending two-vertex certification through zero minors in arbitrary dimension was challenging, with specialist impact in the efficient recognition of structured interval matrix families.

Resolution — 15 October 2026

Resolved affirmatively. Wenqi Zhu (University of Oxford), Kai Zou (King's College London), and Ping Nie (University of Waterloo).

The answer is **yes**. For every positive integer dimension, arbitrary real entrywise bounds, and every common signature, nonsingular sign regularity at the two checker vertices implies nonsingular sign regularity throughout the closed matrix interval. The result allows arbitrary fixed entries, including fixed zeros of both checker parities, and vanishing proper minors. The range $n \geq 1$ includes the original question's $n \geq 5$, with no additional hypothesis on the interval or its signature.

The primary formal result is *IV01.twoCheckerVertices*. Theorem 1.1 of the paper, *Two-vertex certification of nonsingular sign-regular matrix intervals* states the equivalent two-vertex characterization. The Lean proof and verification evidence below gives the exact statement comparison and reproducible formal record. The retained partial-result and literature records below predate this resolution.

Lean proof and verification evidence

Immutable source and dependencies

The proof repository at *22fedcfb6940fd943d67f1582068ed5b9b91a83e* pins Lean ***v4.34.0-rc2*** in its toolchain file and Mathlib ***85e3a25e006c35636f0e53b0e9296caca2685bc0*** in its dependency manifest. The manifest pins all remaining dependencies as well.

Declarations and statement correspondence

All predicates used by the target are defined in *lean/IV01/Definitions.lean*. The complete correspondence compares them with the original catalogue statement:

- **Matrices and dimensions:** *Square n* is *Matrix* (*Fin n*) (*Fin n*) *Real*. The theorem quantifies over every $n \geq 1$ and all real matrices of that dimension. This includes every $n \geq 5$ in the original question.
- **Signature:** *IsSignature n ε* requires $\varepsilon\ k = 1$ or $\varepsilon\ k = -1$ for every $1 \leq k \leq n$. The function $\varepsilon : Nat \to Real$ represents the finite signature by minor order. Unused values are unrestricted; every original signature extends to this representation.
- **Interval:** *EntrywiseLE L U* is the ordinary pointwise inequality, and *InBox L U M* means $L \leq M \leq U$. The interval is closed, with no strict-width condition. Fixed coordinates may occur anywhere.
- **Checker vertices:** *checkerMinus* chooses the lower entry at even row-plus-column parity and the upper entry at odd parity; *checkerPlus* reverses those choices. Lean's zero-based row and column labels preserve the original checker parity when both are shifted to one-based labels.
- **All minors and weak signs:** *SignRegular ε M* quantifies over every order $1 \leq k \leq n$ and every pair of strictly increasing row and column maps. Its inequality is $0 \leq \epsilon_k \det M[I, J]$. These are all minors, with row and column indices in increasing order; proper zero minors are allowed.

- **Nonsingularity and common signature:** *NonsingularSignRegular ε M* is the conjunction of *Matrix.det M ≠ 0* and *SignRegular ε M*. Both checker-vertex assumptions and the conclusion use the same ε. Nonsingularity is assumed only at the two checker vertices and proved for every matrix in the box.

IV01.twoCheckerVertices proves the complete original implication. Its premises are the dimension, signature, entrywise bounds, the two endpoint conditions, and membership of the arbitrary matrix in the box. Its rectangle-exclusion ingredients are proved dependencies, not extra assumptions. There are no additional conditions on fixed coordinates, their directed graph, or vanishing proper minors.

The supporting declarations *IV01.checkerMinus_mem_box* and *IV01.checkerPlus_mem_box* prove that both checker vertices belong to the box. Specializing the universal box property to them gives the converse in the paper's equivalence; the catalogue question itself asks for the implication already proved by the main target. No separate bundled equivalence declaration is claimed.

Reproduction and verification record

The verification record links the dated successful build log, build receipt, and dependency record. On 14 October 2026, all 41 production modules were compiled into initially empty project objects with the pinned compiler and dependency libraries. **The catalogue did not rerun the checks**; the evidence supplied for this status is the external project's verification record. The installed compiler and pinned dependency libraries are disclosed trusted inputs; their original builds were not reconstructed.

With the pinned Lean toolchain installed, reproduce from the immutable revision:

```
git clone https://github.com/erenup/two-vertex-sign-regularity.git
cd two-vertex-sign-regularity
git checkout --detach 22fedcfb6940fd943d67f1582068ed5b9b91a83e
cd lean
lake exe cache get
python3 -B verify.py --output .audit-build
```

The output directory must be new. The Lean guide documents the compiler pin, dependencies, resource requirements and rejection controls. Separate exact rational checks of examples supplement the complete formal theorem; they do not substitute for its universal quantifiers.

Transitive axioms

The literal transitive axiom report for *IV01.twoCheckerVertices* contains only *propext*, *Classical.choice*, and *Quot.sound*. No *sorryAx*, unproved custom axiom or additional trust in native execution supports the target. The verifier checks the source, configuration and dependency pins, compiles fresh project objects, and rejects unexpected axiom output. Its control record records the rejection tests.

The paper-to-Lean map distinguishes the complete main theorem from stronger auxiliary paper formulations and finite example checks.

Further partial resolution — 12 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation.

Theorem G proves the full interval conclusion when the directed graph of nonzero fixed entries is acyclic and fixed zero entries occupy at most one checker parity. Its scaling argument uses Adm–Garloff's Theorem 3.3 with its original attribution. Theorem V proves the full conclusion in dimension five if there are no fixed zeros and every fully fixed adjacent two-by-two block is nonsingular, allowing cyclic fixed graphs. For larger dimensions, Theorem V controls order three only.

The order-two reduction and the nonsingularity/cofactor lemma control orders one, two, and the two highest orders under the original hypotheses. The unrestricted remaining middle-minor signs are not

settled. Neither sufficient condition above covers every interval in the original question, which remains unchanged below.

These partial arguments passed a separate independent Codex AI-agent review. Submission record, verified affiliation and exact certificates. AI assistance is disclosed; finite certificates support examples rather than prove the universal theorem. No historical priority, external human peer review or formal verification is claimed. No Lean verification was performed.

Problem statement

For $\epsilon \in \{-1, 1\}^n$, call a real $n \times n$ matrix M **nonsingular sign regular with signature** ϵ if $\det M \neq 0$ and every minor of order k has determinant d satisfying $\epsilon_k d \geq 0$, for every $1 \leq k \leq n$.

Let $n \geq 5$ and let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise. Define the two checkerboard vertex matrices

$$A^-_{ij} = \begin{cases} L_{ij}, & i+j \text{ even}, \\ U_{ij}, & i+j \text{ odd}, \end{cases} \qquad A^+_{ij} = \begin{cases} U_{ij}, & i+j \text{ even}, \\ L_{ij}, & i+j \text{ odd}. \end{cases}$$

Question

If A^- and A^+ are nonsingular sign regular with the same signature ϵ, must every real matrix M with $L \leq M \leq U$ be nonsingular sign regular with signature ϵ?

The implication would certify a property of an entire interval of matrices from only two of its vertices. Allowing zero minors is essential to the remaining problem.

References

Jürgen Garloff, Mohammad Adm, and Jihad Titi, *A Survey of Classes of Matrices Possessing the Interval Property and Related Properties*, Reliable Computing **22** (2016), 1–14, Conjecture 3.1, p. 7. Adm and Garloff, *Certification of the Sign Regularity of Matrix Intervals*, Acta Scientiarum Mathematicarum, published January 17, 2026, §3, especially the question following Theorem 3.2 and the discussion of admissible signatures.

Earlier status evidence — 2026-09-08

The 2026 survey explicitly retains this question. It records positive results for strictly sign regular matrices, totally nonnegative matrices, tridiagonal matrices, and certain signatures covering all orders at most four. Theorem 3.3 also covers intervals whose fixed entries $L_{ij} = U_{ij}$, if any, all have the same parity of $i + j$; fixed entries of both parities are allowed in the general question. It also records counterexamples when nonsingularity is omitted. Searches for sign-regular interval conjectures and subsequent 2026 results found no resolution of the displayed implication.

Audit update — 2026-09-10

Rechecked the 2026 survey, §3, Theorems 3.2–3.3 and the intervening open question. Tridiagonal and certain signature classes, as well as the specified restrictions on fixed-entry parity, prove substantive subclasses even for $n \geq 5$. The unrestricted nonsingular case remains explicitly open there. Searches for subsequent sign-regular interval certification results found no general resolution; the status now exposes these partial results.